

Homework 10

1. Suppose that a nonnegative random variable X satisfies

$$E(X) = 20.$$

Use Markov's inequality to estimate

$$P(X \geq 40).$$

2. Suppose that a random variable X has mean

$$\mu = 50$$

and variance

$$\sigma^2 = 25.$$

Use Chebyshev's inequality to estimate

$$P(|X - 50| \geq 10).$$

3. A fair coin is tossed 1000 times. Let X denote the number of heads. Use Chebyshev's inequality in the form to estimate the probability that the number of heads differs from 500 by at least 100.

4. Suppose that

$$E(X_i) = 5, \quad \text{Var}(X_i) = 4,$$

and $X_1, \dots, X_{25}$ are independent random variables. Let

$$\bar{X} = \frac{X_1 + \dots + X_{25}}{25}.$$

Find: (a) $E(\bar{X})$ (b) $\text{Var}(\bar{X})$ (c) Using the central limit theorem, approximate $P(\bar{X} < 4.6)$.

5. Suppose that $X_1, \dots, X_{16}$ are independent measurements with

$$E(X_i) = 10, \quad \sigma = 2.$$

Use the central limit theorem to approximate

$$P(|\bar{X} - 10| < 1).$$

6. Suppose that $X_1, \dots, X_{36}$ are independent random variables uniformly distributed on $(0, 1)$.

Using the central limit theorem, approximate

$$P\left(\sum_{i=1}^{36} X_i > 20\right).$$

7. A supermarket records the daily revenue (in thousand dollars). The daily revenues are independent with mean

$$\mu = 50$$

and standard deviation

$$\sigma = 8.$$

Using the central limit theorem, approximate the probability that the average daily revenue over 64 days is greater than 52.